Finoptix Assignment - 3
Fama French Model and MVO Optimization

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Part 1: Data Collection and Preprocessing

The following stocks were used from the S&P 500:

•AXP (American Express)
•IBM
•JPM (JP Morgan)
•MA (Mastercard)
•MCD (McDonald's)
•MS (Morgan Stanley)
•NFLX (Netflix)
•ORCL
•TMUS (T-Mobile US)
•WMT (Walmart)

Start date-> 01-01-2015 End date-> 01-01-2025

Monthly returns were calculated for the above stocks

The following Fama French factors data was also collected from the link given in the assignment pdf-
•Mkt- Rf •HML •SMB •Rf

Part 2: Factor Model Regression:

Through Regression (by OLS), betas and the constant was found out for the fama french factors data, by the below given formula

$$R_p - R_f = \alpha + \beta_m(R_m - R_f) + \beta_{SMB} \cdot \text{SMB} + \beta_{HML} \cdot \text{HML} + \epsilon$$

Results-

	alpha	beta_mkt	beta_smb	beta_hml
AXP	0.002525	1.106261	0.165551	0.508907
IBM	-0.001423	0.858319	-0.218463	0.354909
JPM	0.005823	1.072432	0.126246	0.807738
MA	0.005450	1.090145	-0.420281	0.075363
MCD	0.004487	0.660046	-0.451230	0.193617
MS	0.002633	1.281280	0.365016	0.708539
NFLX	0.012394	1.316241	-0.051501	-0.925597
ORCL	0.001475	1.075806	-0.566883	-0.110075
TMUS	0.011418	0.494402	0.001324	0.016182
WMT	0.003866	0.507864	-0.472635	-0.154550

Interpretation-
beta_smb>0 -> behave like small cap stock
beta_smb<0 -> behaves like large cap stock

beta_hml>0 -> behaves like a value stock (high book to market ratio)
beta_hml<0 -> behaves like a growth stock (low book to market ratio)

Value- Strength of interpretation

Part 3: Factor Neutral Portfolio Construction

In this part we need to assign weights to the different assets in our portfolio such that
$w^\top \beta$ mkt=0
$w^\top \beta$ smb=0
$w^\top \beta$ hml=0

Where, **w1 · β 1,mkt+w2 · β 2,mkt+···+wn · β n,mkt=w⊤ β mkt**
Here w1,w2... are the weights assigned to different asset classes
Since we have the above constraints, we call this a linear constraint optimization problem
We do this so that our portfolio does not tilt towards any other factor

We have created an HML factor neutral portfolio-

```
HML-Neutral Portfolio Weights: {'JPM': 0.0764, 'MA': 0.0865, 'WMT': 0.0569, 'NFLX': 0.1047, 'ORCL': 0.097, 'TMUS': 0.0634, 'IBM': 0.1701, 'MCD': 0.1168,
'MS': 0.1086, 'AXP': 0.1197}
```

Used **CVXPY library for this*

Part 4 : Mean Variance Optimization (MVO)

Used **PyPortfolioOpt** for the following types of MVO

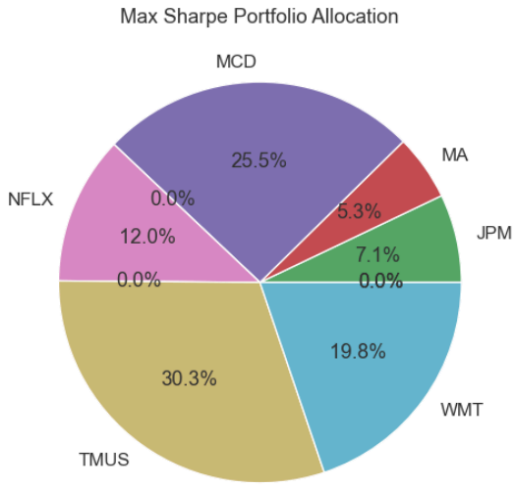
A. Traditional MVO (used mean historical returns)
B. Factor adjusted MVO (used Fama French expected returns)

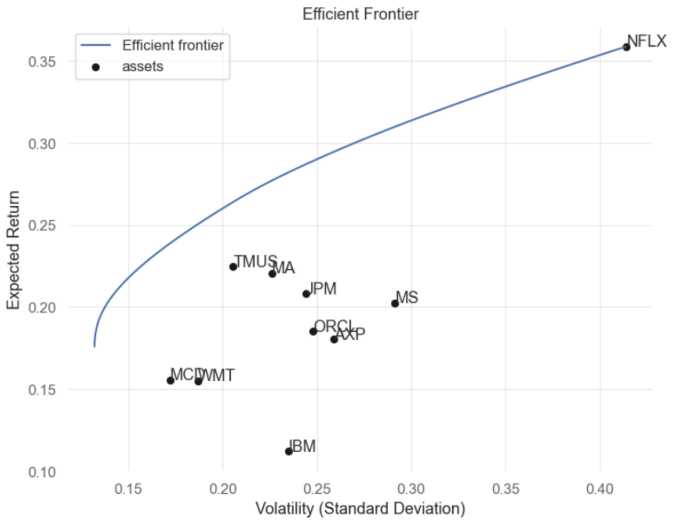
Estimated covariance matrix for all the stocks
Finally created a portfolio to maximise Sharpe Ratio

Results and Visualisations-

A. Traditional MVO

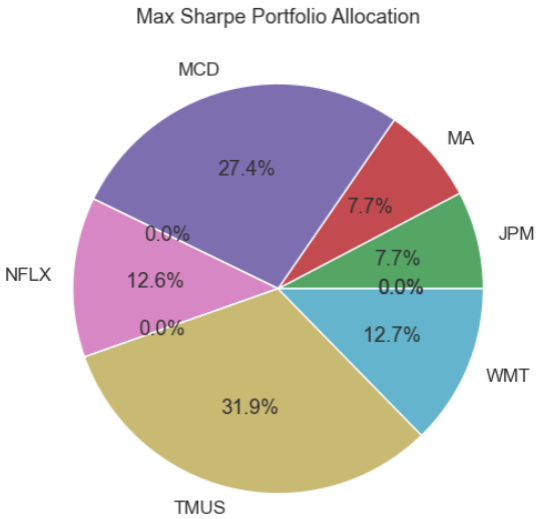
```
Expected annual return: 20.8%
Annual volatility: 14.2%
Sharpe Ratio: 1.46
```

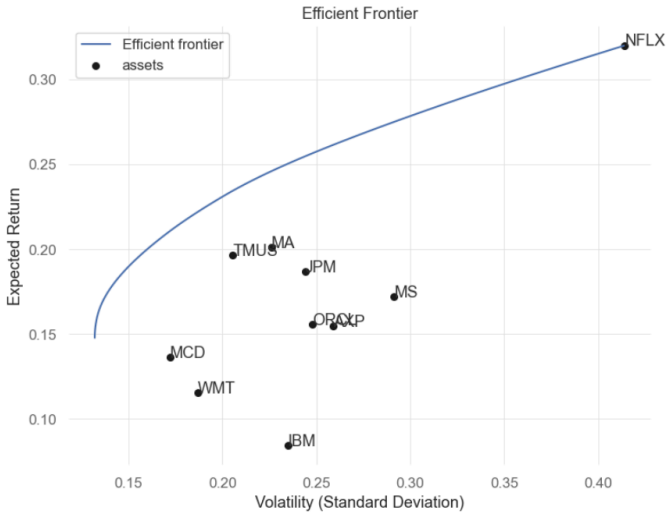


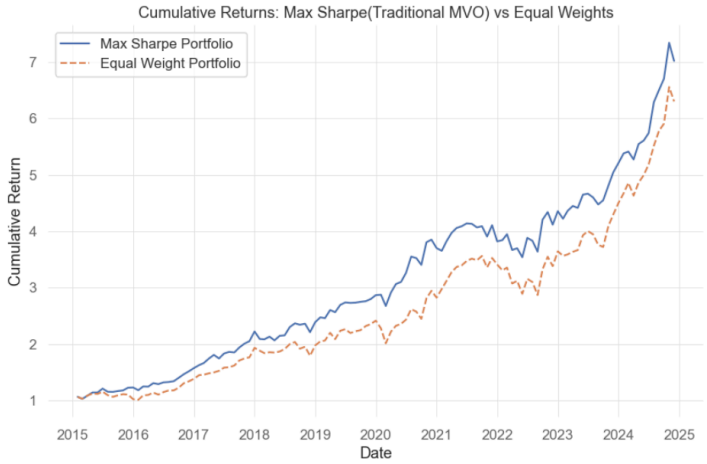


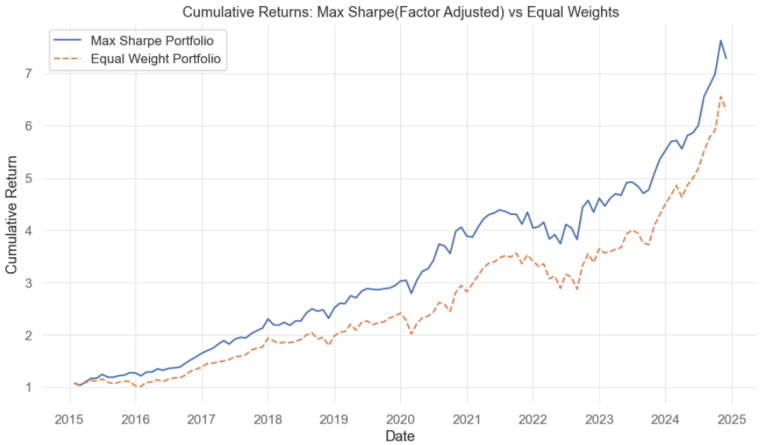
B. Factor Adjusted MVO

```
Expected annual return: 18.5%
Annual volatility: 14.6%
Sharpe Ratio: 1.27
```






Cumulative Returns: Max Sharpe(Traditional MVO) vs Equal Weights
Max Sharpe Portfolio
Equal Weight Portfolio
Cumulative Return
Date
1
2
3
4
5
6
7
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025


Cumulative Returns: Max Sharpe(Factor Adjusted) vs Equal Weights
Max Sharpe Portfolio
Equal Weight Portfolio
Cumulative Return
Date
1
2
3
4
5
6
7
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025